

Question 1 - Instruction Set 1 – x295 - SOLUTION

C. Evaluating our x295 instruction set using Memory Traffic criteria

Table 2

x295 program (1 assembly instruction/ machine code instruction per row)	Fetch (number of memory accesses of word size) + Explain how you obtain your count	Decode/Execute (number of memory accesses of word size) + Explain how you obtain your count
Assembly instruction: ADD x, y, tmp1 Machine code instruction: 0001 <Dest 12 bits> 0000 <Src1 12 bits> 0000 <Src2 12 bits>	Count: 3 Explanation: since the binary encoding of the ADD instruction is 3-word wide	Count: 3 Explanation: since the ADD instruction reads the value of two operands (16 bits each -> 1 word each) from memory and it writes its result (16 bits -> 1 word) to memory
Assembly instruction: SUB x, y, tmp2 Machine code instruction: 0010 <Dest 12 bits> 0000 <Src1 12 bits> 0000 <Src2 12 bits>	Count: 3 Explanation: since the binary encoding of the SUB instruction is 3-word wide	Count: 3 Explanation: since the SUB instruction reads the value of two operands (16 bits each -> 1 word each) from memory and it writes its result (16 bits -> 1 word) to memory

Assembly instruction: MUL tmp1, tmp2, z Machine code instruction: 0011 <Dest 12 bits> 0000 <Src1 12 bits> 0000 <Src2 12 bits>	Count: 3 Explanation: since the binary encoding of the MUL instruction is 3-word wide	Count: 3 Explanation: since the MUL instruction reads the value of two operands (16 bits each -> 1 word each) from memory and it writes its result (16 bits -> 1 word) to memory
Grand Total: 18 memory accesses	Total: 9 memory accesses	Total: 9 memory accesses

Once completed, submit it on [Crowdmark](#) as your answer to Question 1.